

Track 2: The New Way of Working (Sandbox Labs)

 by Ssooraj

Session 1: Prompt Engineering Fundamentals (2 Hours)

Objective: Master the core principles of writing effective prompts—transforming vague business needs into precise AI instructions.

Design Rationale: Prompt engineering is the "new literacy" of the AI era—not about learning what AI can do, but learning how to ask.

Time	Module	Content	Activity Type
0–10 min	Welcome & Review	Recap of Track 1 key takeaways; Lab objectives; Sandbox environment login	Lecture
10–25 min	Why Prompt Engineering Matters	"Bad Prompt vs. Good Prompt" comparison exercise; Show how the same question yields wildly different results	Demo + Discussion
25–45 min	The CO-STAR Framework	Structured prompt framework: C ontext + O bjective + S tyl + T one + A udience + R esponse Format	Lecture + Template Distribution
45–75 min	Hands-On: Build Your First Prompt	Participants write prompts using real Novatech scenarios: • Sales: <i>"Draft a follow-up email for [Client X] about [Product Y]"</i> • Finance: <i>"Extract PO number, amount, and vendor name from this invoice"</i> • Engineering: <i>"Explain this code snippet in plain English"</i>	Individual Practice + Facilitator Roaming
75–90 min	Share & Feedback	3–4 volunteers share their prompts and AI outputs; Group discusses what worked and what didn't	Showcase + Group Critique
90–120 min	Iteration Practice + Wrap-up	Refine prompts based on feedback; Key takeaways; Preview of next session	Practice + Lecture

Deliverable: Each participant leaves with a **"My Personal Prompt Library"** template (Notion/OneNote) containing at least 5 optimized prompts for their specific role.

Session 2: RAG Knowledge Hub in Action (2 Hours)

Objective: Learn to use Novatech's enterprise RAG Knowledge Hub—searching policies, historical documents, and SOPs—while validating AI output accuracy.

Design Rationale: RAG (Retrieval-Augmented Generation) is the "killer app" for enterprise AI—enabling AI to answer questions based on the company's *own* data, not generic internet knowledge.

Time	Module	Content	Activity Type
0–10 min	Welcome & Review	Recap of Session 1; Brief RAG concept introduction ("AI + Company Knowledge = Accurate Answers")	Lecture
10–25 min	RAG Knowledge Hub Demo	Facilitator demonstrates how to: • Upload documents (PDF, Word, Excel) • Search policies ("<i>What is our VPN reset process?</i>") • Verify citations against source documents	Live Demo
25–35 min	Security & Compliance Reminder	What can be uploaded vs. what cannot; PII handling rules; Audit trails	Lecture
35–75 min	Hands-On: Solve Real Business Problems	Participants use the RAG Hub to answer real questions: • HR: "<i>What is the new employee equipment request process?</i>" • Legal: "<i>What is the standard confidentiality clause in our NDAs?</i>" • Operations: "<i>What is the latest version of SOP-042 (Procurement Approval)?</i>" • Engineering: "<i>What is the deployment checklist for production?</i>"	Individual Practice + Facilitator Guidance
75–90 min	The "Fact-Check" Challenge	Facilitator gives a "trap question" (e.g., asking for information not in any document); Participants learn to identify hallucinations and verify sources	Group Challenge
90–110 min	Building Your Personal Knowledge Base	Each participant uploads 3–5 of their most frequently referenced documents to the RAG Hub and tests searches	Practice
110–120 min	Wrap-up & Preview	Key takeaways; FAQs; Preview of next session topic	Lecture

Deliverable: Each participant builds a **personal RAG Knowledge Collection** (3–5 most-used documents now indexed and searchable).

Critical Reminder: Emphasize "**Always Verify Sources**" —the RAG Hub provides answers *and* source document links. If the AI cannot cite a source, the answer cannot be trusted.

Session 3: Building Your AI Assistant (No-Code) (2 Hours)

Objective: Learn to build custom AI assistants/agents without writing code—targeting specific repetitive tasks in your daily workflow.

Design Rationale: No-code AI assistants are "democratized AI"—enabling business users to automate their own workflows without depending on engineering teams.

Time	Module	Content	Activity Type
0–10 min	Welcome & Review	Recap of Session 2; What is an "AI Assistant" vs. a generic chatbot	Lecture
10–25 min	Assistant Builder Demo	Facilitator demonstrates building a custom assistant: • Define assistant's role and task • Upload background knowledge • Set response format and constraints • Test and iterate	Live Demo
25–35 min	Use Case Brainstorm	Participants list their top 3 most repetitive, time-consuming tasks —candidates for assistants	Individual Brainstorm + Sharing
35–75 min	Hands-On: Build Your First Assistant	Participants choose from scenarios: • Sales Assistant: Drafts proposal outlines from client info • HR Assistant: Answers employee FAQs about benefits policies • Finance Assistant: Explains invoice status and payment terms • IT Assistant: Guides employees through password reset or VPN setup • Custom: Participant's own workflow	Individual/Group Practice + Facilitator Guidance
75–90 min	Testing & Iteration	Challenge your own assistant with 5–10 test questions; Refine instructions based on results	Testing + Optimization
90–110 min	Showcase Session ("Show & Tell")	3–4 volunteers showcase their assistants; Group provides feedback and suggestions	Showcase + Feedback
110–120 min	Wrap-up & Preview	Assistant deployment considerations; Preview of next session topic	Lecture

Deliverable: Each participant builds **one functional, task-specific AI assistant** they can use in their daily workflow.

Session 4: AI Workflow Integration & "Human-in-the-Loop" (2 Hours)

Objective: Learn how to integrate AI assistants into daily tools (Slack, Teams, CRM, Jira) and establish "human-in-the-loop" quality control processes.

Design Rationale: AI tools only generate value when embedded into daily workflows. This session focuses on adoption and governance—not just how to use AI, but how to use it responsibly within workflows.

Time	Module	Content	Activity Type
0–10 min	Welcome & Review	Recap of Session 3; Session goal: Moving from "experimentation" to "production-ready"	Lecture
10–25 min	Workflow Integration Demo	Facilitator demonstrates AI integrated into: • Slack/Teams ("/ask" command to query Knowledge Hub) • Jira (Auto-generate release notes) • Salesforce/CRM (Auto-summarize customer interactions)	Live Demo
25–40 min	The "Human-in-the-Loop" Framework	4-Step Decision Framework: 1. Authority: Do I have permission to input this data into AI? 2. Privacy: Does this data contain PII? 3. Accuracy: Is the AI output reasonable? Can I verify it? 4. Oversight: Who should review this before I send/approve?	Lecture + Role Play
40–70 min	Hands-On: End-to-End Workflow Simulation	Participants complete a full "AI-assisted workflow": • Scenario A (Sales): Receive RFP → AI drafts response → Human reviews & edits → Send to client • Scenario B (Finance): Receive invoice → AI extracts data → Human verifies → Enter into ERP • Scenario C (Support): Receive ticket → AI suggests response → Human approves → Send	Group Simulation + Facilitator Guidance
70–85 min	"Failure Scenario" Drill	Facilitator presents scenarios where AI makes a mistake; Participants practice when to override AI and how to escalate	Scenario Drill
85–100 min	Action Planning	Each participant creates a Personal AI Adoption Plan: • Which workflow will I automate? • What do I need to do in the next 30 days? • Whose approval/support do I need?	Individual Planning + Sharing
100–120 min	Graduation Ceremony & Certification	Program wrap-up; Q&A; Certificate presentation: "Novatech AI Champion"; Next steps (Track 3 coming soon)	Graduation Ceremony

Deliverables:

- Each participant creates a **30-day Personal AI Adoption Plan**
- Each participant receives a **"Novatech AI Champion"** certification badge
- A dedicated **Slack/Teams channel** is created for Champions to continue sharing discoveries

Sandbox Environment Specifications

Component	Specification
Platform	Enterprise-grade AI sandbox (e.g., Azure AI Studio, AWS Bedrock Sandbox, or Pluralsight AI Sandbox)
Available Models	2–3 LLMs for comparative experimentation (e.g., Claude 3.5 Sonnet, GPT-4o, Gemini 1.5)
Pre-Configured Data	Novatech sample data (anonymized invoices, policy documents, code snippets, RFP templates)
Security Controls	Zero-retention PII masking; Full audit logging; No production data connections
Access	SSO integration; Individual sandbox instances per participant

Success Metrics (Measured at Track 2 Completion)

Metric	Target
Participant Completion Rate	≥ 90% (40–45 of 50 Champions complete all 4 sessions)
Prompt Engineering Competency	Each participant has ≥ 5 optimized prompts in personal library
RAG Adoption Rate	Each participant has ≥ 3 documents indexed in personal RAG collection
Assistant Building	Each participant has built ≥ 1 functional AI assistant
Participant Satisfaction (NPS)	≥ +70
Self-Reported Time Savings	Each Champion identifies ≥ 1 task with ≥ 2 hours/week automation potential

Summary: Track 2's "Unicorn" Value

Typical "AI Training"	This Track 2 Sandbox Lab
Watch a facilitator demonstrate	Hands-on practice with real data
Generic examples (<i>"Write a poem"</i>)	Novatech-specific scenarios (Invoices, RFPs, SOPs)
No follow-through	Actionable deliverables (Prompt library, RAG collection, Assistant)
One-off event	4-week progressive learning journey
No certification	"AI Champion" certification + Ongoing community
No behavior change	30-day Personal Adoption Plan

Track 2's Promise is Simple:

"40 Champions × 2 hours saved per week = 80 hours/week = 4,000+ hours/year of recovered productivity. And that's just the beginning."